Project Design Phase-I
Solution Architecture

Date	—	**Team ID**	—
Project Name	Global Fuel Prices & Taxation (2020–2026)	**Maximum Marks**	4 Marks

The solution architecture for this project describes how the fuel-price dataset moves from an Excel source into Tableau, where it is prepared, analyzed and presented as visual insights. The design keeps the workflow straightforward and uses the features available in Tableau Desktop.

- Connect Tableau Desktop to the supplied Excel workbook.
- Check field types and geographic roles, and use average (AVG) aggregation for fuel-price comparisons.
- Build worksheets for the global petrol map, regional fuel prices, trends, income-level comparison and taxation analysis.
- Combine the views into the Global Fuel Overview dashboard, with Region and Income Level filters, and present the findings through a Tableau Story.

Solution Architecture Diagram:

The next page shows the data-to-dashboard workflow used in the project.

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data Source	Contains country, region, date, income level, subsidy level, Brent crude value, petrol, diesel, LPG and tax percentage fields.	Excel workbook (.xlsx)
Data Connection	Connects Tableau to the source workbook so its fields can be used in worksheets and dashboards.	Tableau Desktop data connection
Data Preparation	Review field types, confirm Date is recognized as a date, and assign Country a geographic role for the map.	Tableau Data Source page and field settings
Worksheet Layer	Contains eight views for map, regional comparisons, fuel-price trends, income-level comparison and taxation analysis.	Tableau worksheets
Analysis / BI Layer	Uses AVG aggregation for price comparisons, plus sorting, labels, tooltips and dashboard filters to support exploration.	Tableau built-in aggregation and dashboard controls

Component Description Table (Continued)

Component Name	Description / Role in Architecture	Technology Used
Visualization / Reporting	Presents the analysis as a global petrol-price map, regional fuel-price bar charts, a 2020–2026 trend line, a taxation comparison and a taxation-versus-petrol scatter plot.	Tableau charts and Global Fuel Overview dashboard
Users / Audience	Students, researchers and other interested users can compare regions and income groups and explore differences in fuel prices and taxation. The dashboard supports descriptive analysis, not causal proof.	Tableau dashboard and Story

Implementation Notes

- Source: the supplied Excel dataset; the project does not currently require a separate SQL/MySQL database.
- Fuel-price comparisons use average values where appropriate, rather than summing repeated observations.
- The conclusions depend on the coverage and quality of the available dataset; the visualizations do not prove that taxation causes price changes.